UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ServiceNow, Inc.
Petitioner

v.

BMC Software, Inc.
Patent Owner

U.S. Patent No. 7,617,073
Filing Date: February 28, 2003
Issue Date: November 10, 2009

TITLE: SYSTEM AND METHOD FOR ASSESSING AND INDICATING THE HEALTH OF COMPONENTS

DECLARATION OF TAL LAVIAN, PH.D.

Inter Partes Review No. 2015-__

ServiceNow's Exhibit No. 1002

Table of Contents

Page

I. BRIEF SUMMARY OF MY OPINIONS 1

II. INTRODUCTION AND QUALIFICATIONS 2

A. Qualifications and Experience 2

B. Materials Considered 6

III. PERSON OF ORDINARY SKILL IN THE ART 8

IV. RELEVANT BACKGROUND 10

A. Network and Systems Management 12

B. Graphically Displaying the Health Status of IT Components 13

V. THE '073 PATENT'S TECHNIQUE FOR ASSESSING AND INDICATING HEALTH OF IT COMPONENTS 16

A. The Specification 16

B. The Claims of the '073 Patent 20

C. Claim Construction 21

1. "IT component" 23

2. "IT subcomponent" 27

3. "processor" 28

VI. APPLICATION OF PRIOR ART TO CLAIMS 1-4 OF THE '073 PATENT 31

A. Summary of Prior Art References Applied in this Declaration 31

1. Lewis (Ex. 1004) 31

B. Claims 1-4 of the '073 Patent are Disclosed by Lewis 36

1. Claim 1 36

(a) "an IT component processor adapted to compute a component health status of the IT component" (Claim 1[a]) 40

(b) "an IT subcomponent processor adapted to compute a subcomponent health status for the at least one IT subcomponent" (Claim 1[b]) 41

ServiceNow's Exhibit No. 1002

Table of Contents
(continued)

Page

(c) "a renderer adapted to display the health status of the IT component by showing a first indicator for the IT component and a second indicator for the at least one IT subcomponent, wherein the first and second indicator are each separately visible at the same time on a single display window of a display unit" (Claim 1[c]) 43

2. Claim 2 47

3. Claim 3 47

4. Claim 4 49

VII. CONCLUSION 53

ServiceNow's Exhibit No. 1002

I, Tal Lavian, Ph.D., declare as follows:

1. I have personal knowledge of the facts stated in this Declaration, and could and would testify to these facts under oath if called upon to do so.

2. I have been retained by counsel for ServiceNow, Inc. ("Petitioner") in this case as an expert in the relevant art.

3. I have been asked to provide my opinions relating to claims 1-4 of U.S. Patent No. 7,617,073 to Jean-Marc Trinon et al. ("the '073 patent"), which I understand is owned by BMC Software, Inc. ("Patent Owner" or "BMC").

I. BRIEF SUMMARY OF MY OPINIONS

4. Claims 1-4 of the '073 patent purport to disclose a system for displaying health status indications of IT components and subcomponents. They do not describe anything that was new or non-obvious by March 2002, the earliest date listed on the face of the '073 patent. As explained in detail in **Part VI** of this Declaration, the features described in those claims were disclosed and suggested in International Patent Application Pub. No. WO 00/72183 A2 to Lundy Lewis, which pre-dates the earliest application filing date. In broad overview, Lewis discloses a system in which software agents monitor operational data and

compute health status of computer-based services and the network components that the services are composed of or depend on. The separate indications of health status for the services and network components are shown graphically within a single display window.

II. INTRODUCTION AND QUALIFICATIONS

A. Qualifications and Experience

5. I possess the knowledge, skills, experience, training and the education to form an expert opinion and testimony in this case. A detailed record of my professional qualifications, including a list of patents and academic and professional publications, is set forth in my curriculum vitae attached to this declaration as **Exhibit A**.

6. I have more than 25 years of experience in the networking, telecommunications, Internet, and software fields. I received a Ph.D. in Computer Science from the University of California at Berkeley in 2006 and obtained a Master's of Science ("M.Sc.") degree in Electrical Engineering from Tel Aviv University, Israel, in 1996. In 1987, I obtained a Bachelor of Science ("B.Sc.") in Mathematics and Computer Science, also from Tel Aviv University.

7. I am currently employed by the University of California at Berkeley and was appointed as a lecturer and Industry Fellow in the Center of Entrepreneurship and Technology ("CET") as part of UC Berkeley College of Engineering. I have been with the University of California at Berkeley since 2000 where I served as Berkeley Industry Fellow, Lecturer, Visiting Scientist, Ph.D. Candidate, and Nortel's Scientist Liaison, where some positions and projects were done concurrently, others sequentially.

8. I have more than 25 years of experience as a scientist, educator and technologist, and much of my experience relates to computer networking technologies. For eleven years from 1996 to 2007, I worked for Bay Networks and Nortel Networks. Bay Networks was in the business of making and selling computer network hardware and software. Nortel Networks acquired Bay Networks in 1998, and I continued to work at Nortel after the acquisition. Throughout my tenure at Bay and Nortel, I held positions including Principal Scientist, Principal Architect, Principal Engineer, Senior Software Engineer, and led the development and research involving a number of networking technologies. I led the efforts of Java technologies at Bay network and Nortel Networks. In

addition, during 1999-2001, I served as the President of the Silicon Valley Java User Group with over 800 active members from many companies in the Silicon Valley.

9. Prior to that, from 1994 to 1995, I worked as a software engineer and team leader for Aptel Communications, designing and developing mobile wireless devices and network software products. From 1990 to 1993, I worked as a software engineer and team leader at Scitex Ltd., where I developed system and network communications tools (mostly in C and C++).

10. I have extensive experience in the area of network communications and Internet technologies including design and implementation of computer-based systems for managing communications networks, including the ability to monitor and provision networks. While with Nortel Networks and Bay Networks (mentioned above) my work involved the research and development of these technologies. For example, I wrote software for Bay Networks and Nortel Networks Web based network management for Bay Networks switches. I developed Simple Network Management Protocol (SNMP) software for Bay Network switches and software interfaces for Bay Networks' Optivity Network

Management System. I wrote software for Java based device management including software interface to the device management and network management for the Accelar routing switch family network management system.

11. I have extensive experience in network communications, including control and management of routing and switching architectures and protocols in layers 1-7 of the OSI model. Much of my work for Nortel Networks (mentioned above) involved the research and development of network communications technologies. For example, I wrote software for Bay Networks and Nortel Networks switches and routers, developed network technologies for the Accelar 8600 family of switches and routers, the OPTera 3500 SONET switches, the OPTera 5000 DWDM family, and the Alteon L4-7 switching product family. In my lab, I installed, configured, managed and tested many network communications equipment of competitors such as Cisco Systems, Juniper Networks, Extreme Networks, Lucent and Alcatel.

12. I am named as a co-inventor on more than 80 issued patents and I co-authored more than 25 scientific publications, journal articles, and peer-reviewed papers. Furthermore, I am a Senior Member of the Institute of Electrical

and Electronics Engineers ("IEEE").

13. I currently serve as a Principal Scientist at my company Telecomm Net Consulting Inc., where I develop network communication technologies and provide research and consulting in advanced technologies, mainly in computer networking and Internet technologies. In addition, I serve as a Co-Founder and Chief Technology Officer (CTO) of VisuMenu, Inc., where I design and develop architecture of visual IVR technologies for smartphones and wireless mobile devices in the area of network communications. The system is based on cloud networking and cloud computing utilizing Amazon Web Services.

14. Additional details of my background are set forth in my curriculum vitae, attached as **Exhibit A** to this Declaration, which provides a more complete description of my educational background and work experience. I am being compensated for the time I have spent on this matter. My compensation does not depend in any way upon the outcome of this proceeding. I hold no interest in the Petitioner (ServiceNow, Inc.) or the patent owner (BMC Software, Inc.).

B. Materials Considered

15. The analysis that I provide in this Declaration is based on my

education and experience in the field of computer systems and networks, as well as the documents I have considered including U.S. Patent No. 7,617,073 ("'073 patent") **[Ex. 1001]**, which states on its face that it issued from an application filed on February 28, 2003 and claims priority to a provisional application filed on March 1, 2002 ('073, 1:5-7).

16. I reviewed various documents dated prior to March 2002 describing the state of the art at the time of the alleged invention of the '073 patent. As explained below, some of these documents are relied upon as actually disclosing the limitations of the '073 patent, while others are being relied upon primarily for background purposes. The prior art document that I rely upon in this Declaration as actually disclosing the limitations of the claims is:

Exhibit No.	Description of Document
1004	International Patent Application Pub. No. WO 00/72183 A2 to Lundy Lewis

This Declaration also cites the following additional prior art documents for purposes of describing the relevant technology, including the relevant state of the art at the time of the alleged invention of the '073 patent:

Exhibit No.	Description of Document
1005	RFC 1067, A Simple Network Management Protocol (August 1988)
1006	*Excerpts from* Jill Huntington-Lee et al., HP OpenView: A Manager's Guide (1997)

III. PERSON OF ORDINARY SKILL IN THE ART

17. I understand that an assessment of claims of the '073 patent should be undertaken from the perspective of a person of ordinary skill in the art as of the earliest claimed priority date, which I understand is March 1, 2002 (the date of the earliest application listed on the face of the '073 patent).

18. As I explain in more detail in **Parts IV-V** below, the '073 patent relates generally to network and systems management, and more specifically to tools for visually assessing the health of information technology ("IT") components. (*E.g.*, '073, 1:14-20 ("The present innovation preferably relates to Availability Management and Business Processes/Systems Management of the Enterprise Systems Management ('ESM') field . . ."), 2:55-57.) This field of technology was well developed by 2002. In fact, in the "Background of the Invention," the '073 patent references the widely-used Simple Network Management Protocol (SNMP) and several commercially available products,

including Hewlett-Packard OpenView, BMC Patrol and Tivoli Business Systems Manager. ('073, 2:2-5, 2:32-44.)

19. In my opinion, a person of ordinary skill in the art as of March 2002 would possess at least a bachelor's degree in electrical engineering or computer science (or equivalent degree or experience) with at least four years of computer programming experience. Such a person would also have had an understanding of computer systems and networks, the software for monitoring the operation and performance of those systems and networks, such as Network Management Systems (NMS) and Business Processes/Systems Management software, and the tools for creating such software.

20. My opinions regarding the level of ordinary skill in the art are based on, among other things, my experience in the field as described in **Part II** above, and my understanding of the basic qualifications that would be relevant to an engineer or computer programmer tasked with creating a software program for monitoring a computer system and/or network.

21. Although my qualifications and experience exceed those of the hypothetical person having ordinary skill in the art defined above, my analysis and

opinions regarding the ’073 patent have been based on the perspective of a person of ordinary skill in the art as of March 2002.

IV. RELEVANT BACKGROUND

22. The ’073 patent, entitled “System and Method for Assessing and Indicating the Health of Components,” pertains generally to network and systems management and more specifically to a system for visually indicating the health status of information technology (“IT”) components, such as computers, computer programs and networking equipment. (’073, Abstract, 1:14-31, 2:55-57.)

23. The Abstract of the ’073 patent recites:

> A system and method for **visualization of the components** of an enterprise system and the rendering of information about the **health or status** of the enterprise system, **its components, and/or its subcomponents**. The invention uses a **combination of color codes or other indicators** and a combination of algorithms and/or rules-based systems to control the computation of status/severities to associate to components and setup the **color codes and indicators**.

(’073, Abstract (emphasis added).)

24. The ’073 patent further explains::

> The following provides **a visualization of the components** of such systems and the rendering of information about their **“health” or status**. The invention comprises using a **combination of color codes or other indicators** and a combination of algorithms and/ or rules-based systems to control the computation of status/severities to associate to components and setup the **color codes and indicators**. Though the present disclosure focuses on the use of two or three different indicators, it is envisioned that the concepts herein may be applied to a plurality of indicators. The invention remedies the **disadvantages of using a single color code or indicator for providing feedback on the health/status or components** in a complex Enterprise System.

(’073, 2:55-67 (emphasis added).)

25. Thus, in a very high level and simplistic way, the ’073 patent describes visualizing the “health” status of IT components and subcomponents at the same time on a single display window using color codes (e.g., red, yellow green) or other indicators.

26. In this section, I provide a brief background of the state of network

and systems management prior to March 2002 pertinent to the ’073 patent.

A. Network and Systems Management

27. By the 1980s, businesses and enterprises frequently used computers that were connected to a network (commonly known as a “Local Area Network” (LAN)), which allowed the computers to communicate with each other using a network communications technology such as Ethernet. As the number of devices connected to a network increased, so did the need for software tools to monitor and manage those devices. An entire field referred to as “network management” grew out of this need. For example, in August 1988, a standardized protocol known as the Simple Network Management Protocol (“SNMP”) was introduced and remains in wide use today. (*See* RFC 1067, A Simple Network Management Protocol (August 1988), Ex. 1005, § 1.) SNMP, as its name suggests, provides a simple protocol that allows one “network element” (such as a computer) to gather information about and monitor another network element. The protocol provides a means by which “management information for a network element may be inspected or altered by logically remote users.” (*Id.*, § 1.) Each network element contains a software program known as an “agent” that is responsible for

responding to SNMP requests from the network management "station." The agent can, for example, gather information about the network element and report that information back to the management station. (*Id.*, § 3.)

28. SNMP has undergone many enhancements and expansions since its introduction in the 1980s, and was widely used by the time the application for the '073 patent was filed. SNMP was widely used, for example, to monitor the health status of a system and its components, such as whether a particular component is available and operating normally or has experienced a fault.

B. Graphically Displaying the Health Status of IT Components

29. Although SNMP and similar technologies provide techniques to exchange health status information between computers, that status information is not particularly useful unless it can be presented to the operator in a clear and understandable way. Industry responded by introducing software programs that could provide graphical displays to show the components of the network and the current health of those components. The '073 patent acknowledges the existence of these prior art products including OpenView by Hewlett-Packard, and Business Systems Manager by Tivoli. ('073, 2:32-44.) OpenView and Business Systems

Manager were two of the more popular software programs that were frequently used by the network administrators. Computer Associates (Unicenter TNG), PLATINUM Technology (ServerVision) and Bay Networks (Optivity), to name a few others, were also actively developing and providing solutions for network management long before the year 2002.

30. By 2002, it was a common practice for software programs to display the health status of IT components and subcomponents through various colors, such as red, yellow and green. Oftentimes, an indication in red color meant alarm or fault, yellow meant some indication to check the device, while green indicated a good status of the IT components. Moreover, these programs alerted the administrator when a problem occurred on the network by changing the colors of nodes or devices on a network map.

31. For example, by no later than 1997, more than five years before the application for the ’073 patent was filed, HP OpenView Network Node Manager (NMM) offered a graphical display that showed representations of system components and used colors to indicate the health of each component. (*E.g.*, Jill Huntington-Lee et al., *HP OpenView: A Manager’s Guide* (1997), Ex. 1006, p.94

("NNM [Network Node Manager] collects information on the network devices, and automatically displays the devices in a logical map showing current operating status. . . . <u>One of the most important features of NNM is the ability to alert the user when a problem occurs on the network by changing the colors of the node or nodes on the map</u>.") (underlining added).)

32. I have personal hands-on experience with many of these products and concepts years before 2002. In March 1996, I started working for Bay Networks, and in my first project, I wrote software for network management that used similar concepts. Starting in 1998, I personally used HP OpenView and Bay Networks Optivity. In both programs, the concept of representing "health" status was an integral part. In 2000, I used the network device management of the Accelar 1100 Routing-Switch while at Bay Networks. Thus, from personal experience, I know that by the year 2002, the technology for managing IT components and subcomponents was well developed and very mature, with many tools commercially available in the market from many vendors. It was almost impossible to manage large enterprise systems and networks without such tools.

V. THE '073 PATENT'S TECHNIQUE FOR ASSESSING AND INDICATING HEALTH OF IT COMPONENTS

A. The Specification

33. The '073 patent, entitled "System and Method for Assessing and Indicating the Health of Components," generally describes a system for visualizing the health status of information technology ("IT") components, such as computers, computer programs and networking equipment, that form part of a system. ('073, Abstract, 1:14-31, 2:55-57.)

34. As mentioned in **Part IV** above, the '073 patent acknowledges that systems for visually indicating the health of IT components were known and commercially available. ('073, 1:64-2:5 ("When it comes to render the 'health' or status information about the displayed components, the solutions implemented today are relying on some coloration applied to, around, or behind the icons representing the component. . . . This coloration technique has been used since the first implementation of such software."), 2:31-44 (describing HP OpenView, BMC Patrol and Tivoli Business Systems Manager).)

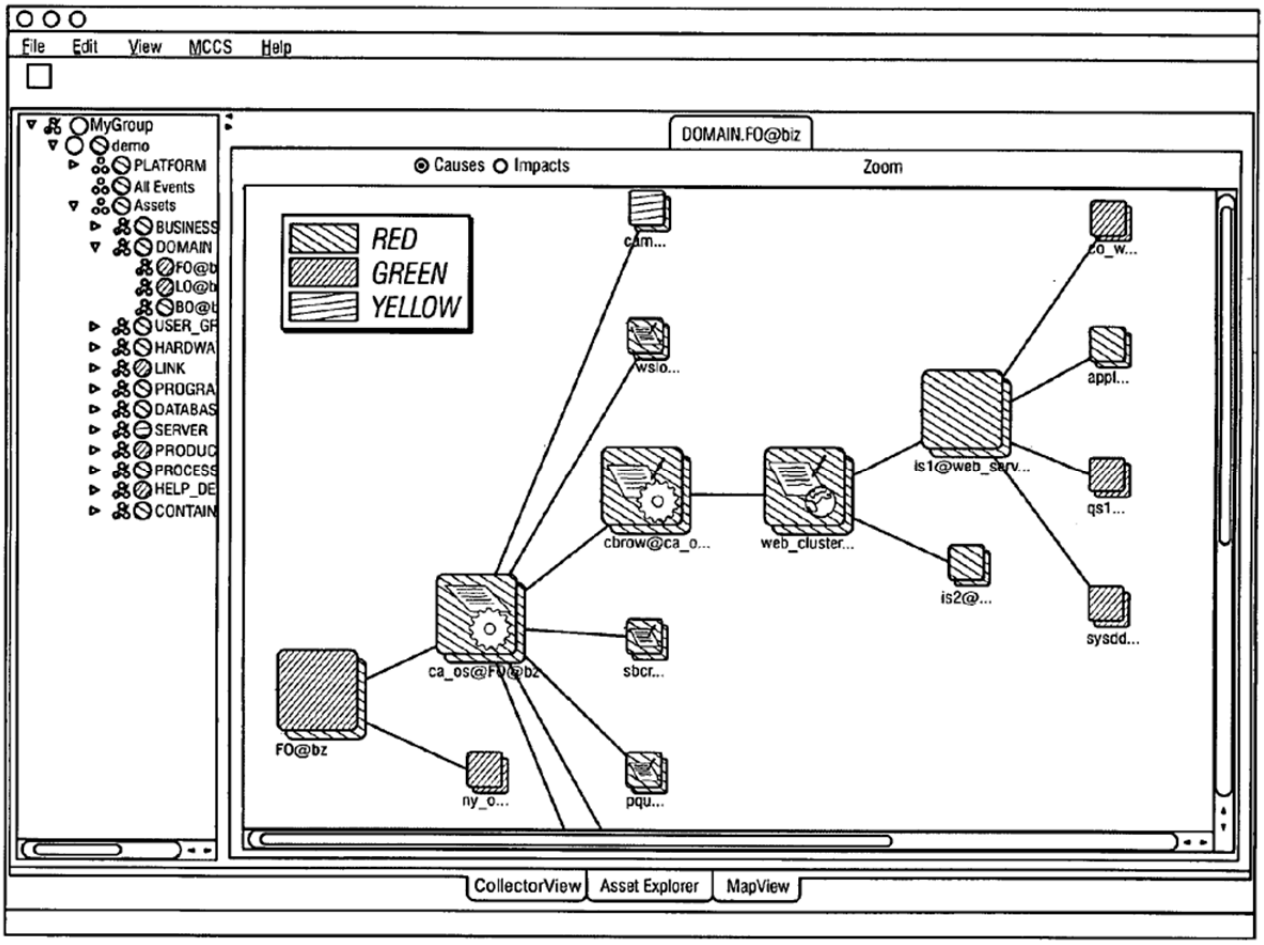
35. The '073 patent asserts, however, that accurately assessing an indicated component health status using prior art systems and methods could

require further manipulation of the graphical interface. ('073, 2:45-51.) For example, existing software could require a user to "drill down" by selectively expanding a hierarchical tree representing the components and subcomponents of the system. ('073, 1:35-52 ("The users can navigate through the representation of the systems by expanding parts of the tree or by selecting the icons representing the component they want to explore further. . . . In doing so, the user can see the components that are part of the selected component or components that the selected component is depending upon."), 2:37-41, 3:14-28 ("These systems and their methods of use require additional navigation through the GUI to distinguish these different situations.").)

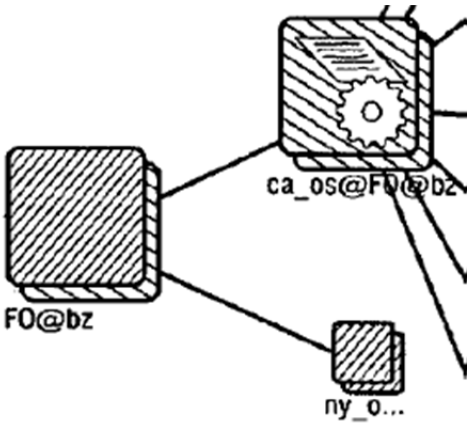
36. Thus, according to the '073 patent, "a need exists to be able to provide more indication information." ('073, 2:49-51.) The '073 patent purports to address that need by disclosing separate <u>visual indications of IT component and</u>

subcomponent health status within a single display window.[1] Figure 1, below, depicts a preferred embodiment of the system:

[1] During prosecution, the applicant asserted that displaying component and subcomponent health status indicators at the same time without requiring a user to navigate or "**drill-down**" was a key difference over the prior art cited by the examiner. (Jan. 12, 2009 Resp. to Office Action, p.11 ("In contrast, the invention of claims 1, 11 and 13 (as amended) provide the health status of an IT component and a subcomponent via a *single* act of rendering - displaying both component and subcomponent health status indicators without requiring the user to perform any affirmative action (i.e., 'navigate'). Ridalfo does not describe, teach or fairly suggest this behavior. In fact, Ridalfo explicitly teaches that the user *must* navigate through plural windows to get such information.") (emphasis in original).)



('073, Fig. 1; *see also id.*, 3:7-8 ("FIG. 1 is an illustrative representation rendered from a preferred embodiment of the system.").)



37. Figure 1 shows a single display window that includes graphical representations of the components and subcomponents of a system arranged into a tree structure and branching hierarchy. ('073, 3:65-4:3.) For example, as shown in the excerpt of Figure 1 at right, component FO@biz is depicted as being composed of components ca_os@FO@biz and ny_os@FO@biz. (*See also* '073, 4:13-15.) The graphical representations include colored indicators for a user to assess the health status of the respective components and subcomponents. ('073, 4:15-19.) For example, "a green indicator is used when a component is healthy and a red one when it is not." ('073, 4:12-13.) If a component is indicated as not healthy, a user can review the health status indications of its subcomponents in order to more fully assess the situation. ('073, 4:20-5:52.)

B. The Claims of the '073 Patent

38. The independent claim addressed in this Declaration—claim 1—purports to recite a system for indicating the health status of one or more IT

components and subcomponents:

1. A system for indicating the health status of an IT component and at least one IT subcomponent comprising:

[a] an IT component processor adapted to compute a component health status of the IT component;

[b] an IT subcomponent processor adapted to compute a subcomponent health status for the at least one IT subcomponent; and

[c] a renderer adapted to display the health status of the IT component by showing a first indicator for the IT component and a second indicator for the at least one IT subcomponent, wherein the first and second indicator are each separately visible at the same time on a single display window of a display unit.

('073, 7:9-22 (Claim 1).) I added the bracketed notations (e.g., "[a]," "[b]," etc.) to facilitate identification of these limitations in my Declaration. The additional claims addressed in this Declaration—i.e., claims 2-4—depend from independent claim 1. I address those claims in detail in **Part VI** below.

C. Claim Construction

39. I have been informed by counsel that invalidity involves a two-step analysis. In the first step, the scope and meaning of a claim is determined by construing the terms of that claim. In the second step, the claim as interpreted is compared to the prior art. Thus, before I address the application of the prior art

to the claims of the ’073 patent in **Part VI** below, I will provide proposed constructions for certain terms in the claims addressed in this Declaration, from the perspective of a person of ordinary skill in the art.

40. I have been informed by counsel that a claim in an unexpired patent subject to *inter partes* review must be given its broadest reasonable construction that is consistent with the specification of the patent in which it appears, which is different from the manner in which the scope of a claim is determined in litigation. I have applied this standard in my analysis below.

41. Before addressing any particular claim language, I note that the specification of the ’073 patent uses broad language in its descriptions of the system. For example:

> Those skilled in the art recognize that the present inventions **not limited to any specific visualization or GUI style**. The inclusion of trees, lists, branches, and/or **any other type** of visualization is to be considered **to be within the scope** of the invention. Moreover, **any indicator** may be composed of **any type of indicia** including, but not limited to particular shapes for color indicators such as background squares, circles or similar shapes, color indicators, icons, icon variations, **or any**

> **type of indicia**.

(’073, 5:53-61 (emphasis added).) The ’073 patent continues:

> Those skilled in the art will recognize that **significant deviation** from this algorithm **is still well within the scope** of the invention. Though certainly **not restrictive** of the embodiments that are considered to be **within the scope** of the invention, the program code incorporated by reference offers one embodiment and those skilled in the art will realize that **significant variation** from this embodiment is meant to be **within the scope of this invention**.

(’073, 5:53-61 (emphasis added).) The ‘073 continues:

> **[V]arious changes, substitutions, and alterations** may be made in the form, construction, and/or arrangement of the elements without departing from **the spirit and scope of the invention . . .**

(’073, 6:65-7:2 (emphasis added).)

1. “IT component”

42. Both independent claims of the ’073 patent recite an “IT component.” In my opinion, the broadest reasonable interpretation of “**IT component**” is a “**part of an information technology system, *i.e.* a system that**

ServiceNow's Exhibit No. 1002

relies at least in part on storing, retrieving, transmitting or manipulating data or information." As explained below, this interpretation is based on the specification and is also consistent with the plain and ordinary meaning to a person of ordinary skill in the art.

43. The written description uses the term "IT" only once, in the Field of the Invention as an acronym for "information technology." ('073, 1:18-19.) Persons of ordinary skill in the art understood "information technology" (or "IT") to refer generally to computer and telecommunications technology involved in storing, retrieving, transmitting or manipulating data or information.

44. The word "component" to persons of ordinary skill in the art is generally understood as referring to a part of a larger system. The specification provides a number of examples of components including hardware, software and "virtual" components that make up a business system:

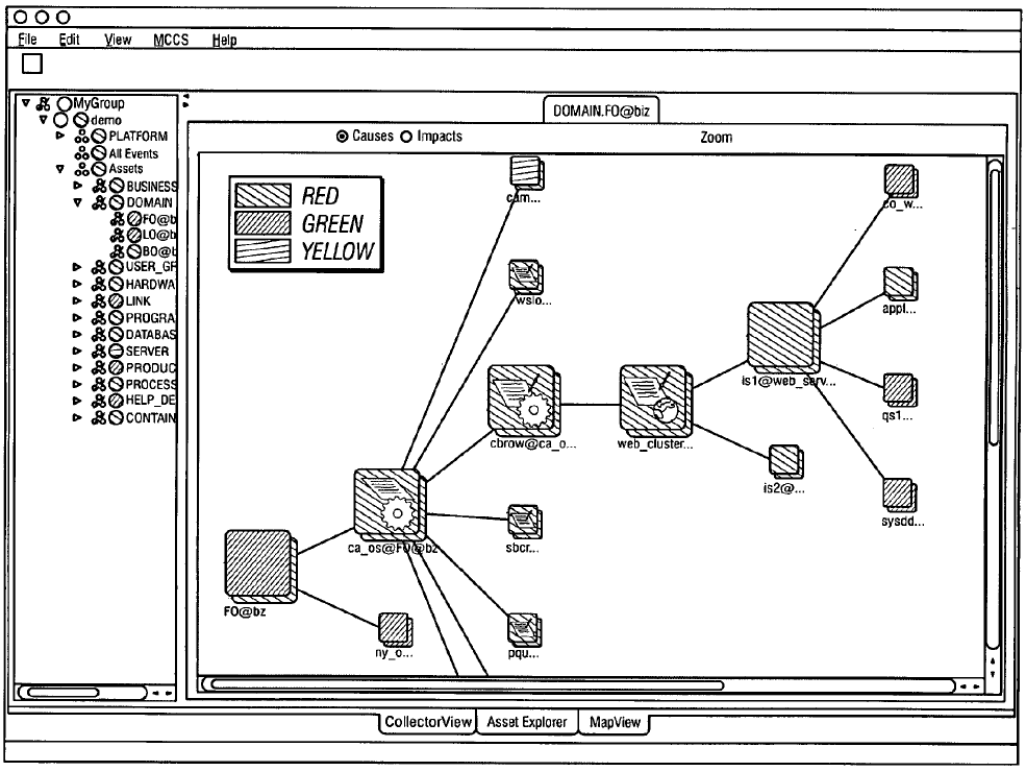
> In today's environments, companies are setting up very complex systems in order to provide services to their customers. <u>Those services are often built using a large variety of components</u>. For example, those <u>components can be computers, computer peripherals, computer programs,</u>

ServiceNow's Exhibit No. 1002

> networking equipment, and manufacturing equipment. Components can also be virtual components like business processes that can be combined into a business system.

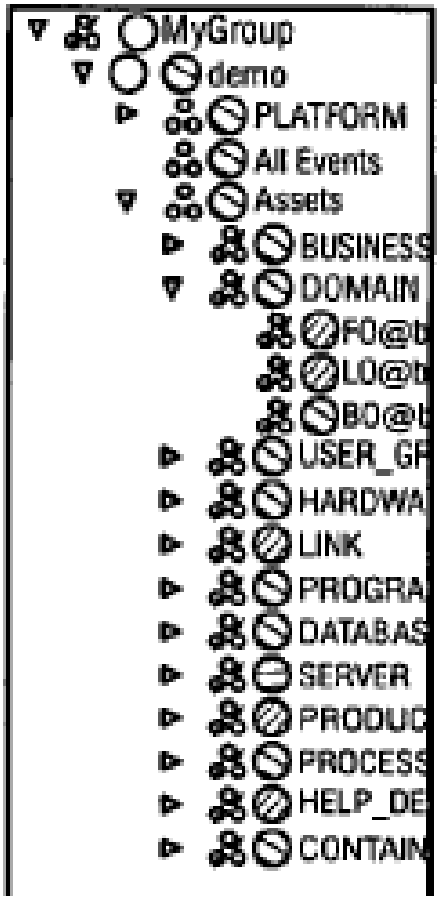
('073, 1:24-31 (underlining added).)

45. With respect to the types of things that can qualify as "IT components," Figure 1 provides additional examples:



('073, Fig. 1.) "Referring to Figure 1, a preferred embodiment of a representation from the system is shown." ('073, 3:65-66.) The left side box of Figure 1 shows a hierarchical listing of components. As explained in the '073 patent: "In this

figure, the <u>components</u> are shown in both a <u>tree structure</u> and a branching hierarchy." ('073, 4:1-3 (underlining added).)



46. I have enlarged this "tree structure" from Figure 1 (shown on the right) to enable easier viewing of the names of the components. As shown, the components include virtual components such as "DOMAIN," and "DATABASE," among others. One of ordinary skill in the art would have understood that a "database," for example, is not a discrete software or hardware component but a service built from a combination of hardware and software components. A "domain" generally refers to a logical grouping of items such as networked computing devices. Some of the names of other components appear to be obscured but likely refer to "USER_GROUP" and "HELP_DESK," which would appear to represent other examples of "virtual" components.

47. As noted previously, the specification is clear that the term "IT

component" can include "virtual components like business processes." ('073, 1:30.) Based on the textual description and examples in figures, one of ordinary skill in the art would have understood that the term "IT component" under its broadest reasonable construction is not limited to a discrete or self-contained hardware or software module, but may include virtual components made up of a number of other components. The broadest reasonable construction of "**IT component**" is therefore a "**part of an information technology system, *i.e.* a system that relies at least in part on storing, retrieving, transmitting or manipulating data or information**."

2. "IT subcomponent"

48. Both independent claims of the '073 patent recite an "IT subcomponent." The specification generally refers to an IT subcomponent as an IT component that is part of, or used by, another IT component. As explained in the Background of the Invention, "[t]he different components relate to each other in different ways. Some components are being parts of others, while some components are using the service provided by other components in some way." ('073, 1:32-35.) The specification goes on to provide an example in Figure 1 in

which "component A is composed of components B and C or depends on components B and C." ('073, 3:16-18.)

49. These disclosures indicate that an "IT subcomponent" is a part (logically or physically) of another IT component. Giving this term its broadest reasonable construction in light of its description in the written description, therefore, an "IT subcomponent" is an "**IT component (a) that is part of another IT component, or (b) that is used or depended upon by another IT component**."

3. "processor"

50. The word "processor" appears only in the claims of the '073 patent, which recite an "IT component processor" and an "IT subcomponent processor" which calculate the health status, respectively, of at least one IT component and at least one IT subcomponent. The word "processor" does not appear anywhere in the written description of the '073 patent. In my opinion, the word "processor" should be understood, under its broadest reasonable construction, as simply "**any combination of software and/or hardware**."

51. Persons of ordinary skill in the art often used "processor" to refer to a particular hardware component that interprets and executes instructions, such

as a microprocessor or CPU. But I do not believe such a narrow meaning is appropriate in the context of the '073 patent. The function of the claimed "processor" is to compute the health status of an IT component or IT subcomponent, and the claims and specification indicate that this function may be carried out through algorithms implemented in software. For example, dependent claim 8 recites that "each IT component processor and each IT subcomponent processor comprise an algorithm to control the computation of the health status of the IT components and IT subcomponents." ('073, Claim 8.) The specification further explains that the technique of the alleged invention "enables the user to have very complex algorithms or rules-based systems that derive the health/status of a component from the events/alerts received on the component itself (if any) and the events/alerts received for its subcomponents or components depending from it." ('073, 3:41-46.)

52. One of ordinary skill in the art would have understood that off-the-shelf processors could not have performed the algorithms or rule-based systems for computing the health status, as described in the '073 patent, without specific software programming. Based on the use of the word "processor" in the claims,

and the corresponding description of the claimed computation function in the written description, I have interpreted "processor" under its broadest reasonable construction as "any combination of hardware and/or software."

53. The following table identifies the above claim constructions:

Claim Term	Broadest Reasonable Construction
"**IT component**"	"part of an information technology system, *i.e.* a system that relies at least in part on storing, retrieving, transmitting or manipulating data or information."
"**IT subcomponent**"	"IT component (a) that is part of another IT component, or (b) that is used or depended upon by another IT component."
"**processor**"	"any combination of software and/or hardware."

VI. APPLICATION OF PRIOR ART TO CLAIMS 1-4 OF THE '073 PATENT

54. I have reviewed and analyzed the prior art references and materials listed in **Part II.B** above. In my opinion, each limitation of claims 1-4 is disclosed by International Patent Application Pub. No. WO 00/72183 A2 to Lundy Lewis **[Ex. 1004]** ("**Lewis**").

55. Counsel has informed me that Lewis properly qualifies as prior art to the '073 patent. Lewis was published on November 30, 2000, which is more than one year before the earliest application filing date on the face of the '073 patent.

56. Before applying Lewis to claims 1-4, I will provide a brief summary of the reference.

A. Summary of Prior Art References Applied in this Declaration

1. Lewis (Ex. 1004)

57. Lewis, entitled "Service Level Management," discloses a system for monitoring the status of computer-based services and their constituent network components, such as a website or Internet access service and the network devices, computers and software that support those services. (Lewis, Ex. 1004, at pp. 1:29-2:3.) As further explained in Lewis:

> A business process (BP) refers to the way(s) in which any type

> of business entity (e.g., company providing goods or services, a department, a university, an ISP, an EC provider, an Internet access provider, nonprofit organization, consultant, etc.) coordinates and organizes work activities and information to produce a valuable commodity. A BP will typically include a number of services, some of which depend on the business's network, and other services which are unrelated to the network. The goal is to <u>identify services which depend on components of the network, and to identify measurable parameters by which accomplishment of the desired services can be monitored and/or controlled</u>.

(Lewis, Ex. 1004, at p. 19:5-12 (underlining added).)

58. Lewis discloses at least two categories of "agents" that are relevant to my discussion below. "An agent, sometimes called a manager, is a software entity that is generally responsive to changes in an environment in which the agency exists." (Lewis, Ex. 1004, p. 21:11-12.) The two types of agents upon which I will focus are "**monitoring agents**" and the "**alarm correlation agent**."

59. Monitoring agents, as their name implies, are responsible for monitoring the activity or status of a particular device. (Lewis, Ex. 1004, p. 39:17-18 ("The monitoring agents **108**-**111** at level-0 identify faults in their areas of

expertise, whereupon they issue control instructions."); Lewis, p. 7:25-26 ("each monitoring agent receiv[es] events of a select type from the network and resolv[es] them into alarms").) Lewis discloses several examples of monitoring agents including "infrastructure agents" that monitor network communications devices such as "bridges, hubs, switches, and routers." (Lewis, Ex. 1004, p. 22:1-3.) Lewis also describes "traffic agents" that monitor flow of data in the network infrastructure, "system agents" that monitor aspects of computer systems such as CPU usage, "application agents" that monitor software applications, among others. (Lewis, p. 22:4-14.)

60. Monitoring agents monitor network components for events. (Lewis, Ex. 1004, p. 5:17-21.) "The event may be generated and transmitted automatically by the network component to an agent monitoring the component, or the agent may poll the network component for the information." (Lewis, Ex. 1004, p. 6:2-4.) Using these events, the agents compute and issue alarms regarding the operation or performance of the components—such as outages or degradation. (Lewis, Ex. 1004, p. 48:9-12 ("Each monitoring agent processes or sifts through its respective detected events and makes a determination about

whether or not to issue an alarm with respect to its area of interest in the enterprise's operation."); Lewis, Ex. 1004, p. 41:23-24 ("Alarms are generated for devices that experience outages and degradation.").)

61. The second type of agent that I will discuss is called an "**alarm correlation agent**." As its name implies, the alarm correlation agent is responsible for "correlating the one or more alarms and determining a state of a service as a function of the correlated alarms." (Lewis, Ex. 1004, p. 10:12-13.) The alarm correlation agent, in other words, determines the status of a service based on the status of its constituent network components. (Lewis, Ex. 1004, p. 8:7-9 ("[A] service is composed of network components and a state of the service depends on the state of the network components."); Lewis, p. 8:15-16 ("[A]n alarm correlation agent [receives] the one or more alarms from the monitoring agents to determine the state of a service.").)

62. Figure 35 of Lewis discloses a graphical user interface (GUI) for displaying the status of services and their components :

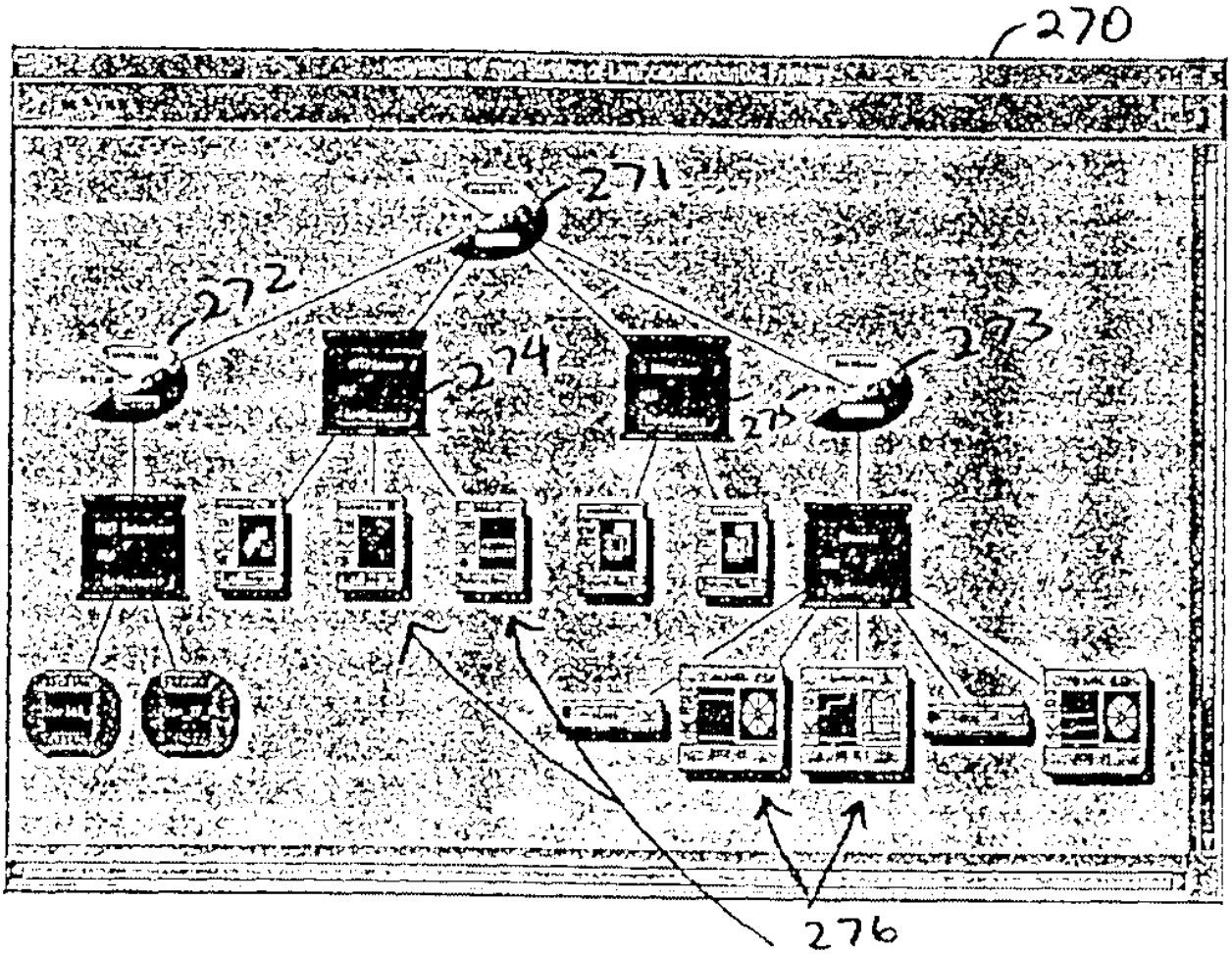


Fig. 35

(Lewis, Ex. 1004, Fig. 35.) The three icons **271**, **272**, **273** at the top of Figure 35, which look like left-facing "Pac-Man" characters, represent services – in particular, ICS Web Site service **271**, Internet Access service **272**, and Backbone service **273**. (Lewis, Ex. 1004, pp. 75:28-76:4.) The top-level ICS Web Site service

271 in the middle "is decomposed into two subservices (Internet access **272** and the backbone **273**), an HTTP daemon **274**, and a Web server **275**." (Lewis, Ex. 1004, p. 76:2-3 (underlining added).) "The light colored icons **276** represent low-level enterprise elements." (Lewis, Ex. 1004, p. 76:4-5 (underlining added).)

63. Lewis further explains that the screen in Figure 35 provides a single display that can show the status of a service and one of its constituent network components. "For example, suppose a BMC patrol agent [*i.e.* a monitoring agent] detects a fault in a server, which in turn affects the service. In this case, both icons might turn red, indicating an alarm." (Lewis, Ex. 1004, p. 76:7-9 (underlining added); *see also* Lewis, p. 44:26-28 (listing "Patrol application agents" among examples of "monitoring agents").)

B. Claims 1-4 of the '073 Patent are Disclosed by Lewis

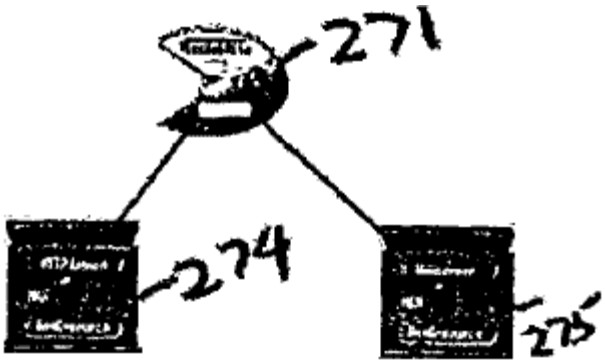
1. Claim 1

64. The preamble of claim 1 recites "[a] system for indicating the health status of an IT component and at least one IT subcomponent." Lewis discloses the preamble of claim 1. As explained in the summary of Lewis provided in **Part VI.A.1** above, Lewis discloses a system for graphically indicating the operational status of

computer-based services and their constituent network components.

65. For purposes of my Declaration, the "**IT component**" in Lewis takes the form of a computer-based service such as the ICS Web Site service **271** in Figure 35, and the "**IT subcomponent**" takes the form of its constituent network components such as Web Server **275**.

66. The excerpt shown at the right shows the exemplary ICS Web Site service **271** that is made up of network components such as, for example, Web Server **275**. (Lewis, Ex. 1004, p. 76:1-3 ("The ICS Web site service **271** is decomposed into . . . an HTTP daemon **274**, and a Web server **275**.").) These components qualify as an "IT component" and "IT subcomponent" as recited in claim 1 of the '073 patent.



67. As I explained in **Part V.C.1** above, the term "**IT component**" should be understood as "part of an information technology system, i.e. a system that relies at least in part on storing, retrieving, transmitting or manipulating data or information." The ICS Web Site service **271** qualifies as an "IT component" under

this definition because it provides a web site service for a larger information technology system. As Lewis explains, "[a] business process (BP) is a way in which a business coordinates and organizes work activities and information to produce a valuable commodity." (Lewis, Ex. 1004, p. 19:26-27.) "A typical BP includes several services, <u>some of those services depend on a network</u>." (Lewis, Ex. 1004, p. 19:29-30 (underlining added).) One of those services, as explained above, is the ICS Web Site Service **271** shown in Figure 35. (Lewis, Ex. 1004, p. 20:11 ("A service is a function that a network provides for the business.").) Because the ICS Web Site service **271** is part of a larger IT system, it qualifies as an "IT component" as claimed.

68. With respect to the claimed "**IT subcomponent**," as I explained in **Part V.C.2** above, that term should be understood as an "IT component (a) that is part of another IT component, or (b) that is used or depended upon by another IT component." The exemplary Web Server **275** qualifies as an "IT subcomponent" under either aspect of this definition. This is because the Web Server **275** is depended upon and part of the larger ICS Web Site service **271**. (Lewis, Ex. 1004, p. 8:7-9 ("[A] service is composed of network components and a state of the

service depends on the state of the network components . . . ."); Lewis, p. 20:13-14 ("Thus, a service may be a function whose performance depends upon performances of network components that support the service.").) As noted above, Lewis discloses that "[t]he ICS Web site service **271** is decomposed into . . . an HTTP daemon **274**, and a Web server **275**." (Lewis, Ex. 1004, p. 76:1-3 (underlining added).) Because the Web Server **275** component forms part of the larger ICS Web Site service **271** (the "IT component"), it qualifies as an "IT subcomponent."

69. Finally, as I will explain in more detail below, Lewis discloses a "system for indicating the health status of an IT component and at least one IT subcomponent," as recited in the preamble. Lewis discloses "a display that indicates a service in the state of a service, where the service is composed of network components and the state of the service depends on the state of the network components." (Lewis, Ex. 1004, p. 12:17-19.) As I will explain below, this display shows (on the same screen) the health status of an IT component (e.g. the ICS Web Site service **271**) and an IT subcomponent (e.g. Web Server **275**). Lewis thus discloses the preamble of claim 1.

(a) "an IT component processor adapted to compute a component health status of the IT component" (Claim 1[a])

70. The **"IT component processor**" in Lewis takes the form of the "**alarm correlation agent**" software that determines the operational status of a computer-based service, such as the ICS Web Service **271** ("IT component").

71. In particular, I explained above in **Part VI.A.1** that Lewis discloses a system in which various "monitoring agents" monitor network components for events and determine whether to issue alarms based on those events. (Lewis, Ex. 1004, p. 48:9-12 ("Each monitoring agent processes or sifts through its respective detected events and makes a determination about whether or not to issue an alarm with respect to its area of interest in the enterprise's operation.").) "An alarm is a message to the overseer that something is wrong, or about to go wrong. Things can go wrong with individual components that make up services." (Lewis, Ex. 1004, p. 26:12-13.) For example, if a device experiences an outage or degradation, an alarm may be issued. (Lewis, Ex. 1004, p. 41:23-24 ("Alarms are generated for devices that experience outages and degradation.").)

72. The "**alarm correlation agent**" (an "IT component processor")

processes these alarms to compute the operational state—i.e., health status—of the service. Lundy explains that "an alarm correlation agent [receives] the one or more alarms from the monitoring agents to determine the state of a service. . . ." (Lewis, Ex. 1004, p. 10:2-3 (underlining added).) "The state of a service may depend, for example, on a collection of service parameter values for availability, reliability, security, integrity and response time." (Lewis, Ex. 1004, p. 20:17-18; *see also* Lewis, p. 48:11-13 ("The issued alarms are sent to the alarm bucket AB for correlation with other alarms, which correlation is performed by the alarm correlation agent ACA."); Lewis, p. 48:19-20 ("The alarms are sent or output to the alarm bucket, step **162**. The alarms are correlated and evaluated to determine the network operation status, step **163**.") (underlining added).) Lewis thus discloses claim 1[a].

(b) "an IT subcomponent processor adapted to compute a subcomponent health status for the at least one IT subcomponent" (Claim 1[b])

73. The "**IT subcomponent processor**" in Lewis takes the form of "monitoring agent" software that determines the operational status of a network component ("IT subcomponent") that forms part of a computer-based service.

The "monitoring agent" software in Lewis computes the "**health status**" of the IT subcomponent by determining whether to issue an alarm indicating that the IT subcomponent has experienced a fault, outage or degradation.

74. In particular, I explained above in **Part VI.A.1** that Lewis discloses a system in which various software agents monitor network components for events. (Lewis, Ex. 1004, p. 5:17-21.) "The event may be generated and transmitted automatically by the network component to an agent monitoring the component, or the agent may poll the network component for the information." (Lewis, Ex. 1004, p. 6:2-4.) Using these events, the agents compute and issue alarms regarding the operation or performance of the components—such as faults, outages or degradation. As explained in Lewis: "<u>Each monitoring agent processes or sifts through its respective detected events</u> <u>and makes a determination about whether or not to issue an alarm</u> with respect to its area of interest in the enterprise's operation." (Lewis, Ex. 1004, p. 48:9-11.) Lewis also makes clear that "alarms" may be issued for components that have experienced a fault, outage or degradation. (Lewis, Ex. 1004, p. 41:23-24 ("Alarms are generated for devices that experience outages and degradation."); *see also* Lewis,

p. 39:17-18 ("As an example, consider fault management. The monitoring agents **108-111** at level-0 identify faults in their areas of expertise, whereupon they issue control instructions.").)

75. Because the monitoring software processes the event data to issue an alarm, the alarm being based on the operational state ("health status") of the component, the monitoring software satisfies claim 1[b].

(c) "a renderer adapted to display the health status of the IT component by showing a first indicator for the IT component and a second indicator for the at least one IT subcomponent, wherein the first and second indicator are each separately visible at the same time on a single display window of a display unit" (Claim 1[c])

76. The "**renderer**" in Lewis takes the form of software that generates a graphical user interface for displaying representations of services, network components and their respective operational status. (Lewis, e.g., pp. 12:9-19, 13:3-7, 14:2-7, 75:28-29.) In particular, Lewis discloses that "Spectrum/ICS" software generates a graphical user interface **270** shown in Figure 35. (Lewis, Ex. 1004, pp. 75:28-29 ("Finally, FIG. 35 shows a simple Spectrum/ICS screen shot **270** of a service decomposed into supporting network devices, computer systems, and

applications.”).) I have reproduced Figure 35 again for convenience:

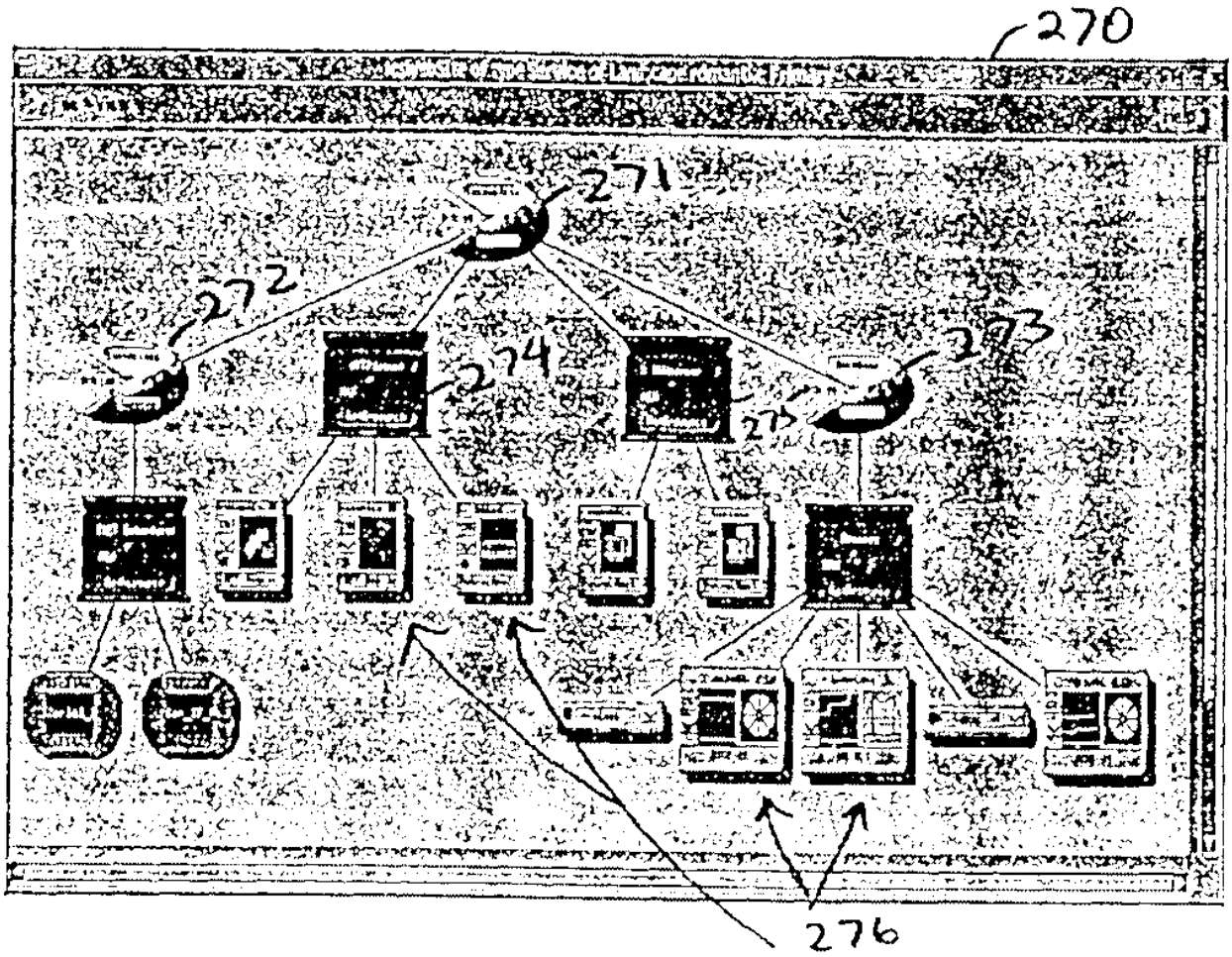


Fig. 35

(Lewis, Ex. 1004, Fig. 35.)

77. The display shown in Figure 35 above satisfies the requirement of claim 1[c] of “display[ing] the health status of the IT component by showing a first

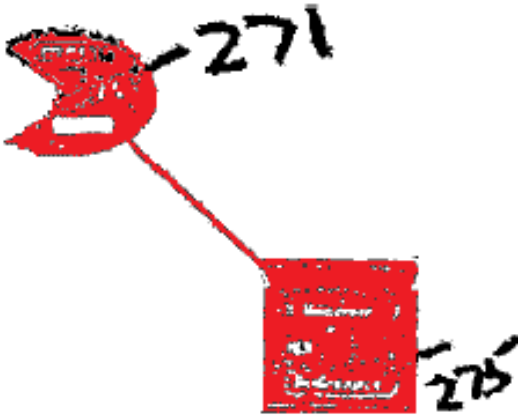
indicator for the IT component and a second indicator for the at least one IT subcomponent," as recited in claim 1**[c]**. The "**first indicator**" in Lewis takes the form of an icon from Figure 35 for a computer-based service, such as the ICS Web Site service **271** shown at the top of the screen. The "**second indicator**" in Lewis takes the form of an icon for a subcomponent of the computer-based service, such as Web Server **275**. The system in Lewis uses these on-screen icons to display the health status of the computer-based service ("IT component"). As explained in Lewis in connection with Figure 35:

> The pull down view menu at the top of FIG. 35 contains a list of possible views and actions (not shown) that can be executed from this console. In addition, the user can click on a particular component and see a list of actions specific to the component. For example, <u>suppose a BMC patrol agent detects a fault in a server, which in turn affects the service. In this case, both icons might turn red, indicating an alarm</u>.

(Lewis, Ex. 1004, p. 76:5-9 (underlining added).)

ServiceNow's Exhibit No. 1002

78. For example, taking the example above, if Web Server **275** in Figure 35 detected a fault that affected ICS Web Site service **271**, then both icons might turn red, indicating an alarm. I have done this coloring of icons in the excerpt from Figure 35 shown at the right.



79. Finally, Lewis discloses that "the first and second indicator are each separately visible at the same time on a single display window of a display unit," as recited in the last part of claim 1[c]. This is shown in Figure 35 above, which as noted in Lewis, is "a graphical user interface screen shot of a service decomposed into supporting network devices, computer systems and applications." (Lewis, Ex. 1004, p. 17:18-19.) For example, the display includes separate icons for Web site service **271** and Web server **275**, and as explained above, the color of each icon indicates the health status of the corresponding service or network component.

80. Thus, for all of the foregoing reasons, Lewis discloses claim 1[c] and, therefore, claim 1.

2. Claim 2

81. Claim 2 depends from independent claim 1 and recites "[t]he system of claim 1 wherein the first indicator and the second indicator are colors." As explained in **Part VI.B.1** above, Lewis discloses each limitation of claim 1. The additional limitations of claim 2 are also disclosed by Lewis.

82. In particular, as explained for claim 1[c] above, the health status of services and network components—i.e., "IT components" and "IT subcomponents"—is visually indicated using separate colored icons. For example, as Lewis explains, if a server experiences a fault, the color of the icon for the server might be turned red, as well as the icon for any service affected by the server fault. (Lewis, Ex. 1004, p. 76:7-9 ("For example, suppose a BMC patrol agent detects a fault in a server, which in turn affects the service. In this case, both icons might turn red, indicating an alarm.").) I see nothing in claim 2, under its broadest reasonable construction, to suggest that the first and second indicators have to be different colors. Lewis thus discloses claim 2.

3. Claim 3

83. Claim 3 depends from independent claim 1 and recites "[t]he system

of claim 1 wherein the IT component processor further comprises a rules-based system to control the computation of the health status of the IT component." As explained in **Part VI.B.1** above, Lewis discloses each limitation of claim 1. Lewis also discloses the additional limitations of claim 3, as explained below.

84. In particular, in my analysis for claim 1[a] above, I explained that Lewis discloses the claimed "IT component processor" in the form of "alarm correlation agent" software. As explained below, Lewis further discloses that the alarm correlation agent can comprise a "rules-based system" that computes the health status of the IT component.

85. In particular, Lewis discloses that the alarm correlation agent can be implemented as a rule-based reasoning (RBR) system. "Rule-based reasoning (RBR) systems, also known as expert systems, production systems, or blackboard systems, generally consist of three basic parts: a working memory, a rule base, and a reasoning algorithm." (Lewis, Ex. 1004, p. 49:16-17.) Lewis specifically discloses that: "<u>An RBR</u> [rule-based reasoning] <u>agent could be used for the alarm correlation agent (ACA)</u>." (Lewis, Ex. 1004, p. 51:14 (underlining added); *see also* Lewis, Ex. 1004, p. 9:8-9 ("The . . . alarm correlation agents may be various

reasoning agents, such as: a rule-based reasoning agent . . . ."); Lewis, p. 49:4-6 ("[T]he alarm correlation agent may implement its analysis of events or alarms using various reasoning paradigms, such as: rule-based reasoning . . . .").) Lewis therefore discloses claim 3.

4. Claim 4

86. Claim 4 depends from independent claim 1 and recites "[t]he system of claim 1 wherein the IT subcomponent processor further comprises a rules-based system to control the computation of the health status of the at least one IT subcomponent." As explained in **Part VI.B.1** above, Lewis discloses each limitation of claim 1. Lewis also discloses the additional limitations of claim 4, as explained below.

87. Claim 4 is similar to claim 3 but instead recites that the "IT <u>subcomponent</u> processor" comprises a rules-based system. In my analysis for claim 1[b] above, I explained that the "monitoring agent" software qualifies as the claimed "IT subcomponent processor." Lewis further discloses that the monitoring agents can be implemented using "rule-based reasoning"—i.e., a "rules-based system." (Lewis, Ex. 1004, p. 49:4-6 ("<u>Each of the</u>

monitoring/mapping agents and the alarm correlation agent may implement its analysis of events or alarms using various reasoning paradigms, such as: rule-based reasoning . . . .") (underlining added).) In fact, Lewis explains that "[m]any vendors provide RBR-based computer monitoring agents, for example BMC Patrol, Tivoli TME, Computer Associates TNG (Islandia, NY), and Platinum ServerVision." (Lewis, Ex. 1004, p. 51:5-6.) Lewis thus discloses claim 4.

88. Finally, in my opinion, the disclosures of Lewis were sufficiently detailed to have allowed a person of ordinary skill in the art to implement the method of claims 1-4 without undue experimentation. As I explained in my discussion of the technology background in **Part IV** above, the technological underpinnings of claims 1-4 were firmly in place by March 2002. The ability to determine and report the status of an IT component was well-known, and in fact, one of the key purposes of the Simple Network Management Protocol (SNMP) introduced in 1988. The '073 patent specifically acknowledges, as prior art, the use of SNMP for this purpose. ('073, 1:61-2:6.) The ability to create a graphical display to show the health status of IT components was known in the prior art, as acknowledged in the '073 patent. ('073, 1:32-60, 2:7-44.) The ability to create

graphical user interfaces using colored icons and other visual indicators was a standard capability of modern windowing operating systems such as Microsoft Windows long before March 2002, and fell within the general knowledge of computer programmers. A person of ordinary skill in the art could therefore have developed a system that practiced claims 1 and 2 based on the disclosures of Lewis in combination with basic computer programming and network management skills, both of which would have been within a skilled artisan's general knowledge.

89. With respect to claims 3 and 4, a person of ordinary skill in the art could have added use of a "rules-based system" to compute the health status of the IT component and IT subcomponent, respectively, based on the disclosures of Lewis. Lewis itself confirms that rule-based agent systems existed long before the '073 patent and were commercially available. (Lewis, Ex. 1004, p. 51:5-6 ("Many vendors provide RBR-based computer monitoring agents, for example BMC Patrol, Tivoli TME. Computer Associates TNG (Islandia, NY), and Platinum ServerVision.").) In fact, Lewis provides far more detail about the structure and operation of a rules-based system than the later-filed '073 patent, which simply

states the possibility of using "rules-based systems" without describing them in any detail. (*Compare* Lewis, Ex. 1004, p. 49:16-50:20 *with* '073, Ex. 1001, 3:41-46.) This is likely because, by the time the '073 patent application was filed, rules-based systems (such as expert systems) were a standard technique for combining rules with facts to perform computer-based decision making, and their basic structure and operation were known to persons of ordinary skill art. And as I noted above, Lewis confirms that "[m]any vendors" had already commercially deployed such systems to compute the health status of IT components or subcomponents. In my opinion, therefore, one of ordinary skill in the art could have implemented a system satisfying each limitation of claims 1-4 based on the disclosures of Lewis without undue experimentation.

VII. CONCLUSION

90. As explained above, claims 1-4 of the ’073 patent describe a system for displaying health status indications of IT components and subcomponents. They do not describe anything that was new or non-obvious by March 2002, the earliest date listed on the face of the ’073 patent. In my opinion, each element of those claims is disclosed or suggested by the prior art described above. As explained in detail above, Lewis discloses a system for monitoring the operational status of services (“IT components”), such as a Web site service, and their constituent network components (“IT subcomponents”), such as network devices, computers and software. The status of services and their network components is computed by software “agents” (“IT component processor” and “IT subcomponent processor”). The agents can be implemented using rules-based reasoning techniques. Agents monitor network components for events, such as operational data. Using the events, the agents compute and issue alarms regarding the operation or performance of the components—such as outages or degradation. The issued alerts are in turn monitored by an “alarm correlation agent” to determine the status of the service. The status of services and their

components is rendered within a single display window by highlighting in red graphical representations of any components and services whose operation is affected by the event(s) that caused the alarm.

91. In signing this Declaration, I recognize that the Declaration will be filed as evidence in a contested case before the Patent Trial and Appeal Board of the United States Patent and Trademark Office. I also recognize that I may be subject to cross-examination in this proceeding. If required, I will appear for cross-examination at the appropriate time. I reserve the right to offer opinions relevant to the invalidity of the '073 patent claims at issue and/or offer testimony in support of this Declaration.

92. I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true, and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under 28 U.S.C. § 1001.

Dated: May 15, 2015

Respectfully submitted,

Tal

Tal Lavian, Ph.D. 5/15/2015

ServiceNow's Exhibit No. 1002

EXHIBIT A

ServiceNow's Exhibit No. 1002

Tal Lavian, Ph.D.

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Research and Consulting: Telecommunications, Network Communications, and Mobile Wireless technologies

- Scientist, educator, and technologist with over 25 years of experience
- Co-author on over 25 scientific publications, journal articles, and peer-reviewed papers
- Named inventor on over 80 issued and filed patents
- Industry fellow and lecturer at UC Berkeley Engineering – Center for Entrepreneurship and Technology (CET)

EDUCATION

- **Ph.D.**, Computer Science specializing in networking and communications, UC Berkeley
- **M.Sc.**, Electrical Engineering, Tel Aviv University
- **B.Sc.**, Mathematics and Computer Science, Tel Aviv University

EXPERTISE

Network communications, telecommunications, Internet protocols and mobile wireless:

- **Communication networks:** Internet Protocols; TCP/IP suite; TCP; UDP; IP; VoIP; Ethernet; network protocols; network software applications; Data Link, Network, and Transport Layers (L2, L3, L4)
- **Internet Software:** Internet software applications; distributed computing; cloud computing; Web applications; FTP; HTTP; Java; C; C++; client server; file transfer; multicast; streaming media
- **Routing/switching:** LAN; WAN; VPN; routing protocols; RIP; BGP; MPLS;OSPF; IS-IS;DNS; QoS; switching; packet switching; network infrastructure; network communication architectures
- **Mobile Wireless:** Wireless LAN; 802.11; cellular systems; mobile devices; smartphone technologies

LITIGATION SUPPORT SERVICES

- Expert witness in numerous USPTO PTAB – Inter Partes Review (IPR) and CBM cases
- Expert witness in Federal courts and the ITC (over 30 cases)
- Expert reports, depositions, and courtroom testimonies
- Skilled articulation of technical material for both technical and non-technical audiences
- Product and technology analysis, patent portfolios, claim charts, patentability research
- Litigation support and technology education in patent disputes
- Past cases involved Cisco, Juniper, HP, Ericsson, Microsoft, Google, Samsung and Apple

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ACCOMPLISHMENTS

- Selected as Principal Investigator for three US Department of Defense (DARPA) projects
- Led research project on networking computation for the US Air Force Research Lab (AFRL)
- Led and developed the first network resource scheduling service for grid computing
- Led wireless research project for an undisclosed US federal agency
- Managed and engineered the first demonstrated transatlantic dynamic allocation of 10Gbs Lambdas as a grid service
- Spearheaded the development of the first demonstrated wire-speed active network on commercial hardware
- Invented over 80 patents; over 50 prosecuted *pro se* in front of the USPTO
- Created and chaired Nortel Networks' EDN Patent Committee
- Current IEEE Senior Member

PROFESSIONAL EXPERIENCE

University of California, Berkeley, Berkeley, CA 2000-Present

Berkeley Industry Fellow, Lecturer, Visiting Scientist, Ph.D. Candidate, Nortel's Scientist Liaison

Some positions and projects were concurrent, others sequential

- Serves as an Industry Fellow and Lecturer at the Center for Entrepreneurship and Technology (CET).
- Studied network services, telecommunication systems and software, communications infrastructure, and data centers
- Developed long-term technology for the enterprise market, integrating communication and computing technologies
- Conducted research projects in data centers (RAD Labs), telecommunication infrastructure (SAHARA), and wireless systems (ICEBERG)
- Acted as scientific liaison between Nortel Research Lab and UC Berkeley, providing tangible value in advanced technologies
- Earned a Ph.D. in Computer Science with a specialization in communications and networking

Telecomm Net Consulting, Inc. (Innovations-IP) Sunnyvale, CA 2006-Present

Principal Scientist

- Consulting in the areas of network communications, telecommunications, Internet protocols, and smartphone mobile wireless devices

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- Providing architecture and system consultation for software projects relating to computer networks, mobile wireless devices, Internet web technologies
- Acting as an expert witness in network communications patent infringement lawsuits

VisuMenu, Inc. – Sunnyvale, CA 2010-Present

Co- Founder and Chief Technology Officer (CTO)

- Design and develop architecture of visual IVR technologies for smartphones and wireless mobile devices in the area of network communications
- Design crawler/spider system for IVR / PBX using Asterisk, SIP and VoIP
- Deploy the system as cloud networking and cloud computing utilizing Amazon Web Services (EC2, S3, VPC, DNS, and RDS)

Ixia, Santa Clara, CA 2008-2008

Communications Consultant

- Researched and developed advanced network communications testing technologies:
  • IxNetwork/IxN2X — tests IP routing and switching devices and broadband access equipment. Provides traffic generation and emulation for the full range of protocols: routing, MPLS, layer 2/3 VPNs, Carrier Ethernet, broadband access, and data center bridging.
  • IxLoad — quickly and accurately models high-volume video, data, and voice subscribers and servers to test real-world performance of multiservice delivery and security platforms.
  • IxCatapult — emulates a broad range of wireless access and core protocols to test wireless components and systems. When combined with IxLoad, provides an end-to-end solution for testing wireless service quality.
  • IxVeriWave — employs a client-centric model to test Wi-Fi and wireless LAN networks by generating repeatable large-scale, real-world test scenarios that are virtually impossible to create by any other means.
  • Test Automation — provides simple, comprehensive lab automation to help test engineering teams create, organize, catalog, and schedule execution of tests.

Nortel Networks, Santa Clara, CA 1996 - 2007

Originally employed by Bay Networks, which was acquired by Nortel Networks

Principal Scientist, Principal Architect, Principal Engineer, Senior Software Engineer

- Held scientific and research roles at Nortel Labs, Bay Architecture Labs, and in the office of the CTO

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Principal Investigator for US Department of Defense (DARPA) Projects

- Conceived, proposed, and completed three research projects: Active Networks, DWDM-RAM, and a networking computation project for Air Force Research Lab (AFRL)
- Led a wireless research project for an undisclosed US federal agency

Academic and Industrial Researcher

- Analyzed new technologies to reduce risks associated with R&D investment
- Spearheaded research collaboration with leading universities and professors at UC Berkeley, Northwestern University, University of Amsterdam, and University of Technology, Sydney
- Evaluated competitive products relative to Nortel's products and technology
- Proactively identified prospective business ideas, which led to new networking products
- Predicted technological trends through researching the technological horizon and academic sphere
- Developed software for switches, routers and network communications devices
- Developed systems and architectures for switches, routers, and network management
- Researched and developed the following projects:
  - Data-Center Communications: network and server orchestration 2006-2007
  - DRAC: SOA-facilitated L1/L2/L3 network dynamic controller 2003-2007
  - Omega: classified wireless project for undisclosed US Federal Agency 2006
  - Open Platform: project for the US Air Force Research Laboratory (AFRL) 2005
  - Network Resource Orchestration for Web Services Workflows 2004-2005
  - Proxy Study between Web/Grids Services and Network Services 2004
  - Streaming Content Replication: real-time A/V media multicast at edge 2003-2004
  - DWDM-RAM: US DARPA-funded program on agile optical transport 2003-2004
  - Packet Capturing and Forwarding Service on IP and Ethernet traffic 2002-2003
  - CO2: content-aware agile networking 2001-2003
  - Active Networks: US DARPA-funded research program 1999-2002
  - ORE: programmable network service platform 1998-2002
  - JVM Platform: Java on network devices 1998-2001
  - Web-Based Device Management: network device management 1996-1997

Technology Innovator and Patent Leader

- Created and chaired Nortel Networks' EDN Patent Committee
- Facilitated continuous stream of innovative ideas and their conversion into intellectual property rights
- Developed intellectual property assets through invention and analysis of existing technology portfolios

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Aptel Communications, Netanya, Israel 1994-1995

Software Engineer, Team Leader

Start-up company focused on mobile wireless CDMA spread spectrum PCN/PCS

- Developed a mobile wireless device using an unlicensed band [Direct Sequence Spread Spectrum (DSSS)]
- Designed and managed a personal communication network (PCN) and personal communication system (PCS), the precursors of short text messages (SMS)
- Designed and developed network communications software products (mainly in C/C++)
- Brought a two-way paging product from concept to development

Scitex Ltd.,Herzeliya, Israel 1990-1993

Software Engineer, Team Leader

Software and hardware company acquired by Hewlett Packard (HP)

- Developed system and network communications (mainly in C/C++)
- Invented Parallel SIMD Architecture
- Participated in the Technology Innovation group

Shalev, Ramat-HaSharon, Israel 1987-1990

Start-up company

Software Engineer

- Developed real-time software and algorithms (mainly in C/C++ and Pascal)

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PROFESSIONAL ASSOCIATIONS

- IEEE Senior Member
- IEEE CNSV co-chair Intellectual Property SIG (2013)
- President Next Step Toastmasters (an advanced TM club in the Silicon Valley) (2013)
- Technical Co-Chair, IEEE Hot Interconnects 2005 at Stanford University
- Member, IEEE Communications Society (COMMSOC)
- Member, IEEE Computer Society
- Member, IEEE Systems, Man, and Cybernetics Society
- Member, IEEE-USA Intellectual Property Committee
- Member, ACM, ACM Special Interest Group on Data Communication (SIGCOM)
- Member, ACM Special Interest Group on Hypertext, Hypermedia and Web (SIGWEB)
- Member, IEEE Consultants' Network (CNSV)
- Global Member, Internet Society (ISOC)
- President Java Users Group – Silicon Valley Mountain View, CA,1999-2000
- Toastmasters International

ADVISORY BOARDS

- Quixey (present) – search engine for wireless mobile apps
- Mytopia – mobile social games
- iLeverage – Israeli Innovations

PROFESSIONAL AWARDS

- Top Talent Award – Nortel
- Top Inventors Award – Nortel EDN
- Certified IEEE-WCET - Wireless Communications Engineering Technologies
- Toastmasters International - Competent Communicator (twice)
- Toastmasters International - Advanced Communicator Bronze

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Patents and Publications

(Not an exhaustive list)

Patents Issued:

- **US 8,688,796** Rating system for determining whether to accept or reject objection raised by user in social network
- **US 8,572,303** Portable universal communication device
- **US 8,553,859** Device and method for providing enhanced telephony
- **US 8,548,131** Systems and methods for communicating with an interactive voice response system
- **US 8,537,989** Device and method for providing enhanced telephony
- **US 8,341,257** Grid proxy architecture for network resources
- **US8,161,139** Method and apparatus for intelligent management of a network element
- **US 8,146,090** Time-value curves to provide dynamic QoS for time sensitive file transfer
- **US 8,078,708** Grid proxy architecture for network resources
- **US 7,944,827** Content-aware dynamic network resource allocation
- **US7,860,999** Distributed computation in network devices
- **US 7,734,748** Method and apparatus for intelligent management of a network element
- **US 7,710,871** Dynamic assignment of traffic classes to a priority queue in a packet forwarding device
- **US 7,580,349** Content-aware dynamic network resource allocation
- **US 7,433,941** Method and apparatus for accessing network information on a network device
- **US 7,359,993** Method and apparatus for interfacing external resources with a network element
- **US 7,313,608** Method and apparatus for using documents written in a markup language to access and configure network elements
- **US 7,260,621** Object-oriented network management interface

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- **US 7,237,012** Method and apparatus for classifying Java remote method invocation transport traffic
- **US 7,127,526** Method and apparatus for dynamically loading and managing software services on a network device
- **US7,047,536** Method and apparatus for classifying remote procedure call transport traffic
- **US7,039,724** Programmable command-line interface API for managing operation of a network device
- **US6,976,054** Method and system for accessing low-level resources in a network device
- **US6,970,943** Routing architecture including a compute plane configured for high-speed processing of packets to provide application layer support
- **US6,950,932** Security association mediator for Java-enabled devices
- **US6,850,989** Method and apparatus for automatically configuring a network switch
- **US6,845,397** Interface method and system for accessing inner layers of a network protocol
- **US6,842,781** Download and processing of a network management application on a network device
- **US6,772,205** Executing applications on a target network device using a proxy network device
- **US6,564,325** Method of and apparatus for providing multi-level security access to system
- **US6,175,868** Method and apparatus for automatically configuring a network switch
- **US6,170,015** Network apparatus with Java co-processor
- **US 8,619,793** Dynamic assignment of traffic classes to a priority queue in a packet forwarding device
- **US 8687,777** Systems and methods for visual presentation and selection of IVR menu
- **US 8,681,951** Systems and methods for visual presentation and selection of IVR menu

- **US 8,625,756** Systems and methods for visual presentation and selection of IVR menu
- **US 8,594,280** Systems and methods for visual presentation and selection of IVR menu
- **US 8,548,135** Systems and methods for visual presentation and selection of IVR menu
- **US 8,406,388** Systems and methods for visual presentation and selection of IVR menu
- **US 8,345,835** Systems and methods for visual presentation and selection of IVR menu
- **US 8,223,931** Systems and methods for visual presentation and selection of IVR menu
- **US 8,160,215** Systems and methods for visual presentation and selection of IVR menu
- **US 8,155,280** Systems and methods for visual presentation and selection of IVR menu
- **US 8,054,952** Systems and methods for visual presentation and selection of IVR menu
- **US 8,000,454** Systems and methods for visual presentation and selection of IVR menu
- **EP 1,905,211** Technique for authenticating network users
- **EP 1,142,213** Dynamic assignment of traffic classes to a priority queue in a packet forwarding device
- **EP 1,671,460** Method and apparatus for scheduling resources on a switched underlay network
- **CA 2,358,525** Dynamic assignment of traffic classes to a priority queue in a packet forwarding device

Patent Applications Published and Pending:
(Not an exhaustive list)

- **US 20140105025** Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device
- **US 20140105012** Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device
- **US 20140012991** Grid Proxy Architecture for Network Resources
- **US 20130080898** Systems and Methods for Electronic Communications
- **US 20130022191** Systems and Methods for Visual Presentation and Selection of IVR Menu
- **US 20130022183** Systems and Methods for Visual Presentation and Selection of IVR Menu
- **US 20130022181** Systems and Methods for Visual Presentation and Selection of IVR Menu
- **US 20120180059** Time-Value Curves to Provide Dynamic QOS for Time Sensitive File Transfers
- **US 20120063574** Systems and Methods for Visual Presentation and Selection of IVR Menu
- **US 20110225330** Portable Universal Communication Device
- **US 20100220616** Optimizing Network Connections
- **US 20100217854** Method and Apparatus for Intelligent Management of a Network Element
- **US 20100146492** Translation of Programming Code
- **US 20100146112** Efficient Communication Techniques
- **US 20100146111** Efficient Communication in a Network
- **US 20090313613** Methods and Apparatus for Automatic Translation of a Computer Program Language Code
- **US 20090313004** Platform-Independent Application Development Framework
- **US 20090279562** Content-aware dynamic network resource allocation
- **US 20080040630** Time-Value Curves to Provide Dynamic QoS for Time Sensitive File Transfers
- **US 20070169171** Technique for authenticating network users
- **US 20060123481** Method and apparatus for network immunization
- **US 20060075042** Extensible Resource Messaging Between User Applications and Network Elements in a Communication Network

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- **US 20050083960** Method and Apparatus for Transporting Parcels of Data Using Network Elements with Network Element Storage
- **US 20050076339** Method and Apparatus for Automated Negotiation for Resources on a Switched Underlay Network
- **US 20050076336** Method and Apparatus for Scheduling Resources on a Switched Underlay Network
- **US 20050076173** Method And Apparatus for Preconditioning Data to Be Transferred on a Switched Underlay Network
- **US 20050076099** Method and Apparatus for Live Streaming Media Replication in a Communication Network
- **US 20050074529** Method and apparatus for transporting visualization information on a switched underlay network
- **US 20040076161** Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device
- **US 20020021701** Dynamic Assignment of Traffic Classes to a Priority Queue in a Packet Forwarding Device
- **WO 2007/008976** Technique for Authenticating Network Users
- **WO 2006/063052** Method and apparatus for network immunization
- **WO2000/0054460** Method and apparatus for accessing network information on a network device

ServiceNow's Exhibit No. 1002

Publications

(Not an exhaustive list)

- "R&D Models for Advanced Development & Corporate Research" Understanding Six Models of Advanced R&D - Ikhlaq Sidhu, Tal Lavian, Victoria Howell - University of California, Berkeley. Accepted paper for 2015 ASEE Annual Conference and Exposition- June 2015
- "Communications Architecture in Support of Grid Computing",Tal Lavian,Scholar's Press 2013ISBN 978-3-639-51098-0.
- "Applications Drive Secure Lightpath Creation across Heterogeneous Domains, Feature Topic Optical Control Planes for Grid Networks: Opportunities, Challenges and the Vision." Gommans L.; Van Oudenaarde B.; Dijkstra F.; De Laat C.; Lavian T.; Monga I.; Taal A.; Travostino F.; Wan A.; IEEE *Communications Magazine*, vol. 44, no. 3, March 2006, pp. 100-106.
- *Lambda Data Grid: Communications Architecture in Support of Grid Computing*. Tal I. Lavian, Randy H. Katz; Doctoral Thesis, University of California at Berkeley. January 2006.
- "Information Switching Networks." Hoang D.B.; T. Lavian;*The 4th Workshop on the Internet, Telecommunications and Signal Processing, WITSP2005*, December 19-21, 2005, Sunshine Coast, Australia.
- "Impact of Grid Computing on Network Operators and HW Vendors." Allcock B.; Arnaud B.; Lavian T.; Papadopoulos P.B.; Hasan M.Z.; Kaplow W.; *IEEE Hot Interconnects at Stanford University 2005*, pp.89-90.
- *DWDM-RAM: AData Intensive Grid Service Architecture Enabled by Dynamic Optical Networks*. Lavian T.; Mambretti J.; Cutrell D.; Cohen H.J; Merrill S.; Durairaj R.; Daspit P.; Monga I.; Naiksatam S.; Figueira S.; Gutierrez D.; Hoang D.B., Travostino F.; CCGRID 2004, pp. 762-764.
- *DWDM-RAM: AnArchitecture for Data Intensive Service Enabled by Next Generation Dynamic Optical Networks*. Hoang D.B.; Cohen H.; Cutrell D.; Figueira S.; Lavian T.; Mambretti J.; Monga I.; Naiksatam S.; Travostino F.; Proceedings IEEE Globecom 2004, Workshop on High-Performance Global Grid Networks, Houston, 29 Nov. to 3 Dec. 2004, pp.400-409.
- *Implementation of a Quality of Service Feedback Control Loop on Programmable Routers*. Nguyen C.; Hoang D.B.; Zhao, I.L.; Lavian, T.; Proceedings, 12th IEEE International Conference on Networks 2004. (ICON 2004) Singapore, Volume 2, 16-19 Nov. 2004, pp.578-582.
- *A Platform for Large-Scale Grid Data Service on Dynamic High-Performance Networks*. Lavian T.; Hoang D.B.; Mambretti J.; Figueira S.; Naiksatam S.; Kaushil N.; Monga I.; Durairaj R.; Cutrell D.; Merrill S.; Cohen H.; Daspit P.; Travostino F; GridNets 2004, San Jose, CA., October 2004.
- *DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks*. Figueira S.; Naiksatam S.; Cohen H.; Cutrell D.; Daspit, P.; Gutierrez D.; Hoang D. B.; Lavian T.; Mambretti J.; Merrill S.; Travostino F; Proceedings, 4th IEEE/ACM International Symposium on Cluster Computing and the Grid, Chicago, USA, April 2004, pp. 707-714.
- *DWDM-RAM: Enabling Grid Services with Dynamic Optical Networks*. Figueira S.; Naiksatam S.; Cohen H.; Cutrell D.; Gutierrez D.; Hoang D.B.; Lavian T.; Mambretti J.; Merrill S.; Travostino F.; 4th IEEE/ACM International Symposium on Cluster Computing and the Grid, Chicago, USA, April 2004.

ServiceNow's Exhibit No. 1002

- *An Extensible, Programmable, Commercial-Grade Platform for Internet Service Architecture*. Lavian T.; Hoang D.B.; Travostino F.; Wang P.Y.; Subramanian S.; Monga I.; IEEE Transactions on Systems, Man, and Cybernetics on Technologies Promoting Computational Intelligence, Openness and Programmability in Networks and Internet Services Volume 34, Issue 1, Feb. 2004, pp.58-68.
- *DWDM-RAM: An Architecture for Data Intensive Service Enabled by Next Generation Dynamic Optical Networks*. Lavian T.; Cutrell D.; Mambretti J.; Weinberger J.; Gutierrez D.; Naiksatam S.; Figueira S.; Hoang D. B.; Supercomputing Conference, SC2003 Igniting Innovation, Phoenix, November 2003.
- *Edge Device Multi-Unicasting for Video Streaming*. Lavian T.; Wang P.; Durairaj R.; Hoang D.; Travostino F.; Telecommunications, 2003. ICT 2003. 10th International Conference on Telecommunications, Tahiti, Volume 2, 23 Feb.-1 March, 2003 pp. 1441-1447.
- The SAHARA Model for Service Composition Across Multiple Providers. Raman B.; Agarwal S.; Chen Y.; Caesar M.; Cui W.; Lai K.; Lavian T.; Machiraju S.; Mao Z. M.; Porter G.; Roscoe T.; Subramanian L.; Suzuki T.; Zhuang S.; Joseph A. D.; Katz Y.H.; Stoica I.; Proceedings of the First International Conference on Pervasive Computing. ACM Pervasive 2002, pp. 1-14.
- *Enabling Active Flow Manipulation in Silicon-Based Network Forwarding Engines*. Lavian T.; Wang P.; Travostino F.; Subramanian S.; Duraraj R.; Hoang D.B.; Sethaput V.; Culler D.; Proceeding of the Active Networks Conference and Exposition, 2002.(DANCE) 29-30 May 2002, pp. 65-76.
- *Practical Active Network Services within Content-Aware Gateways*. Subramanian S.; Wang P.; Durairaj R.; Rasimas J.; Travostino F.; Lavian T.; Hoang D.B.; Proceeding of the DARPA Active Networks Conference and Exposition, 2002.(DANCE) 29-30 May 2002, pp. 344-354.
- *Active Networking on a Programmable Network Platform.* Wang P.Y.; Lavian T.; Duncan R.; Jaeger R.; Fourth IEEE Conference on Open Architectures and Network Programming (OPENARCH), Anchorage, April 2002.
- *Intelligent Network Services through Active Flow Manipulation*. Lavian T.; Wang P.; Travostino F.; Subramanian S.; Hoang D.B.; Sethaput V.; IEEE Intelligent Networks 2001 Workshop (IN2001), Boston, May 2001.
- Intelligent Network Services through Active Flow Manipulation. Lavian T.; Wang P.; Travostino F.; Subramanian S.; Hoang D.B.; Sethaput V.; Intelligent Network Workshop, 2001 IEEE 6-9 May 2001, pp.73 - 82.
- *Enabling Active Flow Manipulation in Silicon-based Network Forwarding Engine*. Lavian, T.; Wang, P.; Travostino, F.; Subramanian S.; Hoang D.B.; Sethaput V.; Culler D.; Journal of Communications and Networks, March 2001, pp.78-87.
- *Active Networking on a Programmable Networking Platform*. Lavian T.; Wang P.Y.; IEEE Open Architectures and Network Programming, 2001, pp. 95-103.
- *Enabling Active Networks Services on a Gigabit Routing Switch*. Wang P.; Jaeger R.; Duncan R.; Lavian T.; Travostino F.; 2nd Workshop on Active Middleware Services, 2000.

ServiceNow's Exhibit No. 1002

- *Dynamic Classification in Silicon-Based Forwarding Engine Environments*. Jaeger R.; Duncan R.; Travostino F.; Lavian T.; Hollingsworth J.; Selected Papers. 10th IEEE Workshop on Metropolitan Area and Local Networks, 1999. 21-24 Nov. 1999, pp.103-109.
- *Open Programmable Architecture for Java-Enabled Network Devices*. Lavian, T.; Jaeger, R. F.; Hollingsworth, J. K.; IEEE Hot Interconnects Stanford University, August 1999, pp. 265-277.
- *Open Java SNMP MIB API*. Rob Duncan, Tal Lavian, Roy Lee, Jason Zhou, Bay Architecture Lab Technical Report TR98-038, December 1998.
- *Java-Based Open Service Interface Architecture*. Lavian T.; Lau S.; BAL TR98-010 Bay Architecture Lab Technical Report, March 1998.
- *Parallel SIMD Architecture for Color Image Processing*. Lavian T. Tel – Aviv University, Tel – Aviv, Israel, November 1995.
- *Grid Network Services, Draft-ggf-ghpn-netservices-1.0*.George Clapp, Tiziana Ferrari, Doan B. Hoang, Gigi Karmous-Edwards, Tal Lavian, Mark J. Leese, Paul Mealor, Inder Monga, Volker Sander, Franco Travostino, Global Grid Forum(GGF).
- *Project DRAC: Creating an applications-aware network.*Travostino F.; Keates R.; Lavian T.; Monga I.; Schofield B.; Nortel Technical Journal, February 2005, pp. 23-26.
- *Optical Network Infrastructure for Grid, Draft-ggf-ghpn-opticalnets-1.*Dimitra Simeonidou, Reza Nejabati, Bill St. Arnaud, Micah Beck, Peter Clarke, Doan B. Hoang, David Hutchison, Gigi Karmous-Edwards, Tal Lavian, Jason Leigh, Joe Mambretti, Volker Sander, John Strand, Franco Travostino, Global Grid Forum(GGF) GHPN Standard GFD-I.036 August 2004.
- *Popeye - Using Fine-grained Network Access Control to Support Mobile Users and Protect Intranet Hosts.*Mike Chen, Barbara Hohlt, Tal Lavian, December 2000.

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Presentations and Talks

(Not an exhaustive list)

- Lambda Data Grid: An Agile Optical Platform for Grid Computing and Data-intensive Applications.
- Web Services and OGSA
- WINER Workflow Integrated Network Resource Orchestration.
- Technology & Society.
- Abundant Bandwidth and how it affects us?
- Active Content Networking(ACN).
- DWDM-RAM:Enabling Grid Services with Dynamic Optical Networks .
- Application-engaged Dynamic Orchestration of Optical Network Resources .
- A Platform for Data Intensive Services Enabled by Next Generation Dynamic Optical Networks .
- Optical Networks.
- Grid Optical Network Service Architecture for Data Intensive Applications.
- Optical Networking & DWDM.
- OptiCal Inc.
- OptiCal & LUMOS Networks.
- Optical Networking Services.
- Business Models for Dynamically Provisioned Optical Networks.
- Business Model Concepts for Dynamically Provisioned Optical Networks.
- Optical Networks Infrastructure.
- Research Challenges in agile optical networks.
- Services and Applications' infrastructure for agile optical networks.
- Impact on Society.
- TeraGrid Communication and Computation.
- Unified Device Management via Java-enabled Network Devices.
- Active Network Node in Silicon-Based L3 Gigabit Routing Switch.
- Active Nets Technology Transfer through High-Performance Network Devices.
- Programmable Network Node: Applications.
- Open Innovation via Java-enabled Network Devices.
- Practical Considerations for Deploying a Java Active Networking Platform.
- Open Java-Based Intelligent Agent Architecture for Adaptive Networking Devices.
- Java SNMP Oplet.
- Open Distributed Networking Intelligence: A New Java Paradigm.
- Open Programmability.
- Active Networking On A Programmable Networking Platform.
- Open Networking through Programmability.
- Open Programmable Architecture for Java-enabled Network Devices.

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- Integrating Active Networking and Commercial-Grade Routing Platforms.
- Programmable Network Devices.
- To be smart or not to be?

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